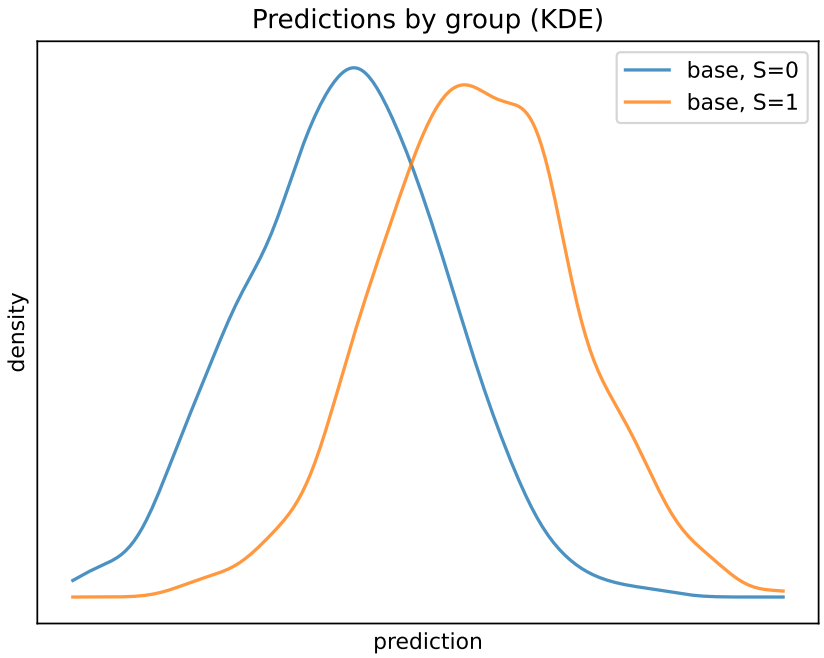
Demographic Parity in regression & classiffication (unawareness)

1 Setting

Data $(X, S, Y) \in \mathcal{X} \times \mathcal{S} \times \mathbb{R}$; $S \in \{1, \dots, K\}$.

- Group weights $p_s := \mathbb{P}(S = s)$, feature laws $\mu_s := Law(X \mid S = s)$, overall $\mu = \sum_s p_s \, \mu_s$.
- Bayes regressor (unconstrained): $\eta(x) := \mathbb{E}[Y \mid X = x]$.
- For $f : \mathcal{X} \to \mathbb{R}$, squared risk $R(f) = \mathbb{E}[(Y - f(X))^2] = \mathbb{E}[(Y - \eta(X))^2] + \mathbb{E}[(\eta(X) - f(X))^2]$.
- Pushforward: $f_{\#}\mu$ is law of $f(X)$ when $X \sim \mu$.



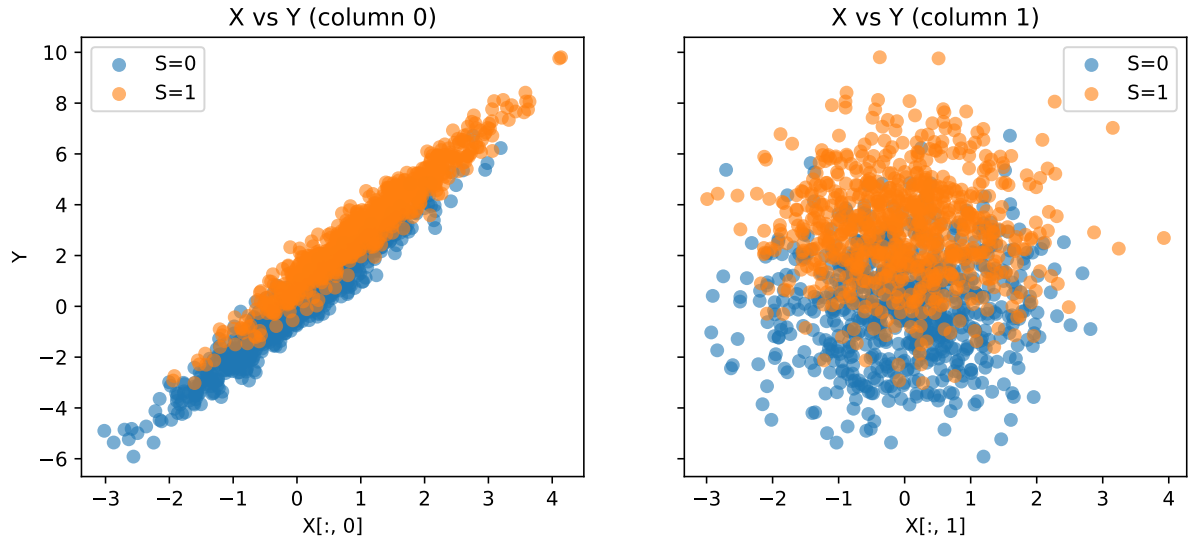
2 DP

DP for regression (unawareness): $f(X) \perp S$, i.e. $f_{\#}\mu_s$ is the same for all s.

$$\min_{f : \mathcal{X} \to \mathbb{R}} \mathbb{E}[(Y - f(X))^2] \quad \text{s.t.} \quad f(X) \perp S.$$

Order preservation (within group): if $\eta(X) < \eta(X')$ and $S = S'$, then $f(X) < f(X')$ a.s.

Key impossibility: if f has DP while η is not DP, then f cannot preserve order (proof by contrapositive: DP+$\uparrow$ order $\Rightarrow$ η DP).



3 Two groups ⇒ reduce to "excess-mass" pieces

Let $\alpha := \mu_1 - \mu_2$ (*signed* measure). Jordan decomposition $\alpha = \alpha^+ - \alpha^-$ with equal mass $m := \alpha^+(\mathcal{X}) = \alpha^-(\mathcal{X}) = \frac{1}{2}\|\mu_1 - \mu_2\|_{\mathrm{TV}}$.

$$\mu_+ := \alpha^+/m, \qquad \mu_- := \alpha^-/m.$$

Supports split $\mathcal{X}$ into $\mathcal{X}_+$ (where group 1 dominates), $\mathcal{X}_-$ (group 2 dominates), and the tie region $\mathcal{X}_=$.

$$\boxed{f \text{ is DP} \iff (f_{\#}\mu_+) = (f_{\#}\mu_-)}$$

(*So DP is governed by how f acts on the disagreement regions; $\mathcal{X}_=$ is irrelevant to DP.*)

3. The 2D lift and the OT cost (this is the main trick)

Pick a reference μ dominating $\mu_\pm$. Define the *signed density gap*

$$\Delta(x) := \begin{cases} \dfrac{\mathrm{d}\mu_+}{\mathrm{d}\mu}(x), & x \in \mathcal{X}_+, \\ -\dfrac{\mathrm{d}\mu_-}{\mathrm{d}\mu}(x), & x \in \mathcal{X}_-, \\ 0, & x \in \mathcal{X}_=, \end{cases} \qquad \text{and note} \quad \mathrm{d}\mu = \frac{1}{|\Delta|}\,\mathrm{d}\mu_\pm \text{ on } \mathcal{X}_\pm.$$

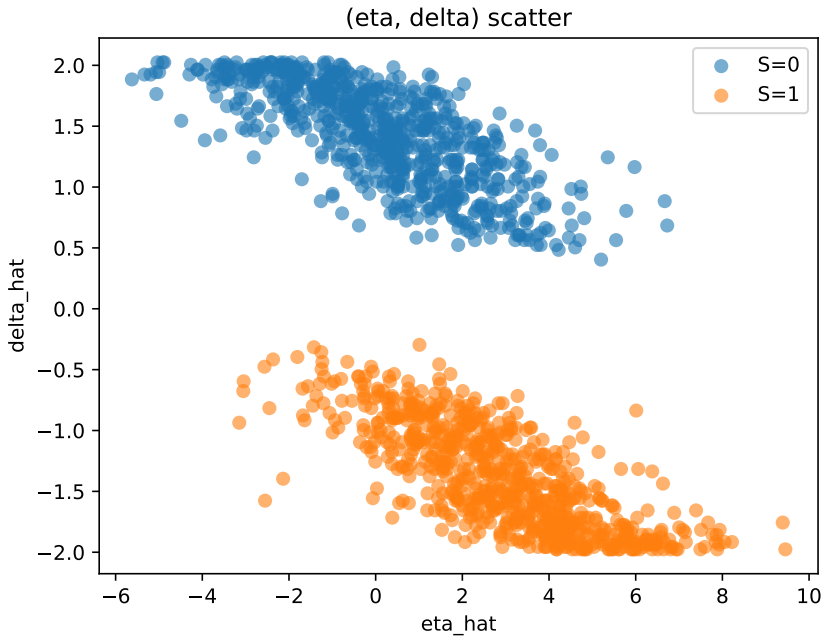
Lift $x \mapsto x^\uparrow := (\eta(x), \Delta(x)) = (h, d) \in \Omega := \{(h, d) \in \mathbb{R}^2 : d \neq 0\}$. Let $\mu_\pm^\uparrow$ be the laws of $x^\uparrow$ under $\mu_\pm$.

Cost (multi→one): $c((h, d), y) = \dfrac{(h - y)^2}{|d|}$.

4. Barycenter program

$$\boxed{\min_{\nu \in \mathcal{P}(\mathbb{R})} \ \mathrm{OT}_c(\mu_+^\uparrow, \nu) + \mathrm{OT}_c(\mu_-^\uparrow, \nu)}$$

This equals the optimal *excess* risk among DP regressors (first reduce risk to $\mathbb{E}[(\eta - f)^2]$, then use $\mathrm{d}\mu = \mathrm{d}\mu_\pm/|\Delta|$ on $\mathcal{X}_\pm$).



4 Collapse to a standard 2-marginal OT

For fixed (x_1, x_2) with $x_i = (h_i, d_i)$,

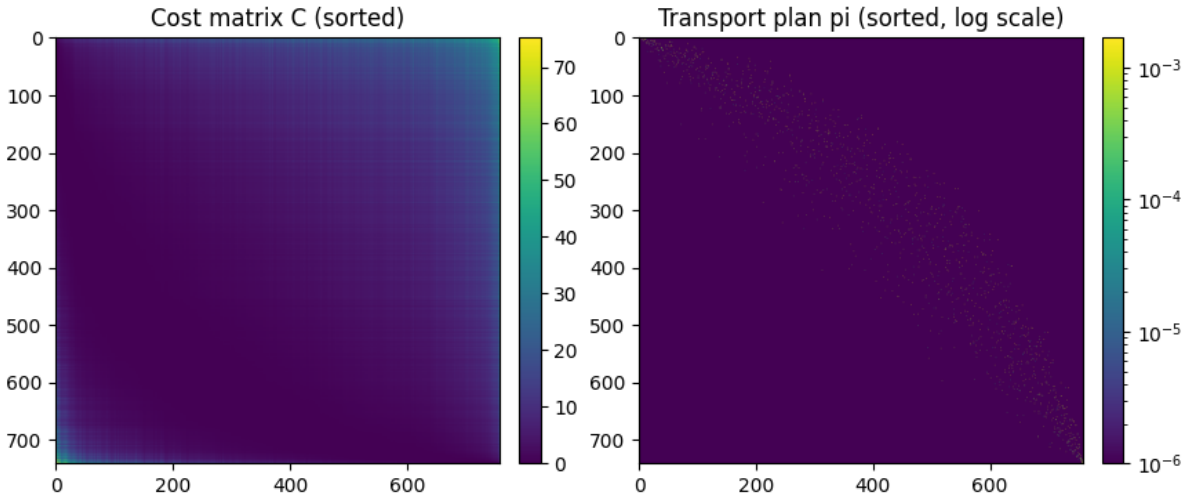
$$\min_y \frac{(h_1 - y)^2}{|d_1|} + \frac{(h_2 - y)^2}{|d_2|} \quad \Rightarrow \quad y^\star = \frac{\frac{h_1}{|d_1|} + \frac{h_2}{|d_2|}}{\frac{1}{|d_1|} + \frac{1}{|d_2|}},$$

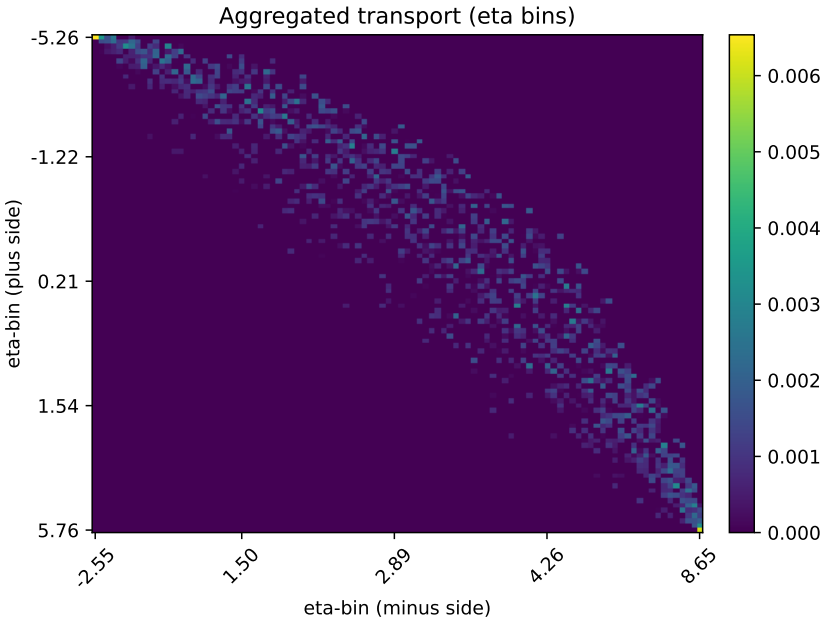
and minimum value

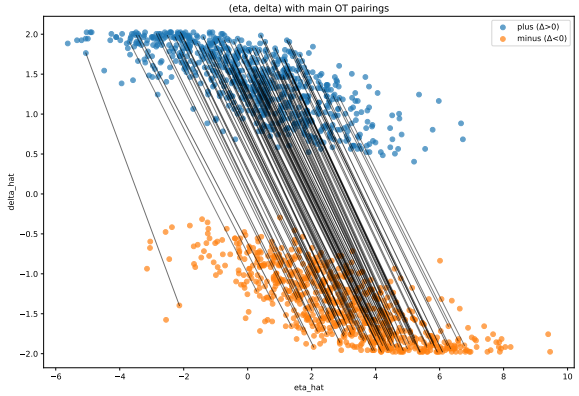
$$\boxed{C((h_1, d_1), (h_2, d_2)) = \frac{(h_2 - h_1)^2}{|d_1| + |d_2|}}.$$

Hence

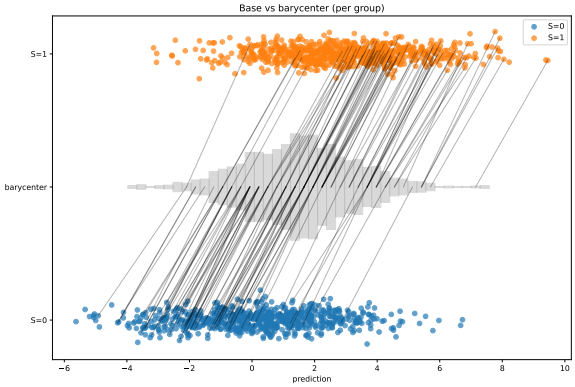
$$\min_\nu \ \mathrm{OT}_c(\mu_+^\uparrow, \nu) + \mathrm{OT}_c(\mu_-^\uparrow, \nu) \ = \ \mathrm{OT}_C(\mu_+^\uparrow, \mu_-^\uparrow).$$







Transport lines from (η, Δ) to the fair prediction (top 30% largest moves; max 150 lines).



Barycenter links from (η, Δ) to the fair prediction (top 30% largest moves; max 150 links).

5 Existence, uniqueness, and maps (the mild regularity)

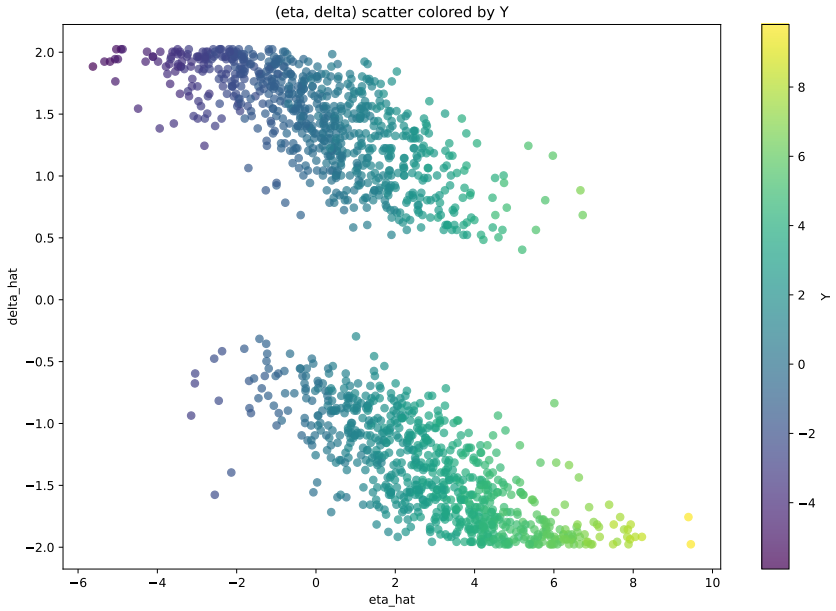
Assumption (zero mass on graphs). $\mu_{\pm}^{\uparrow}$ give zero measure to any set $\{(h, d) : h = F(d),\ d \neq 0\}$. (this reduces to non-atomicity assumption in the aware setting)

Under this and $\mathbb{E}[Y^2] < \infty$:

- The barycenter is *unique*: ν^{bar}.
- Each $\mathrm{OT}_c(\mu_{\pm}^{\uparrow}, \nu^{\mathrm{bar}})$ is realized by a *transport map* $f_{\pm} : \Omega \to \mathbb{R}$.

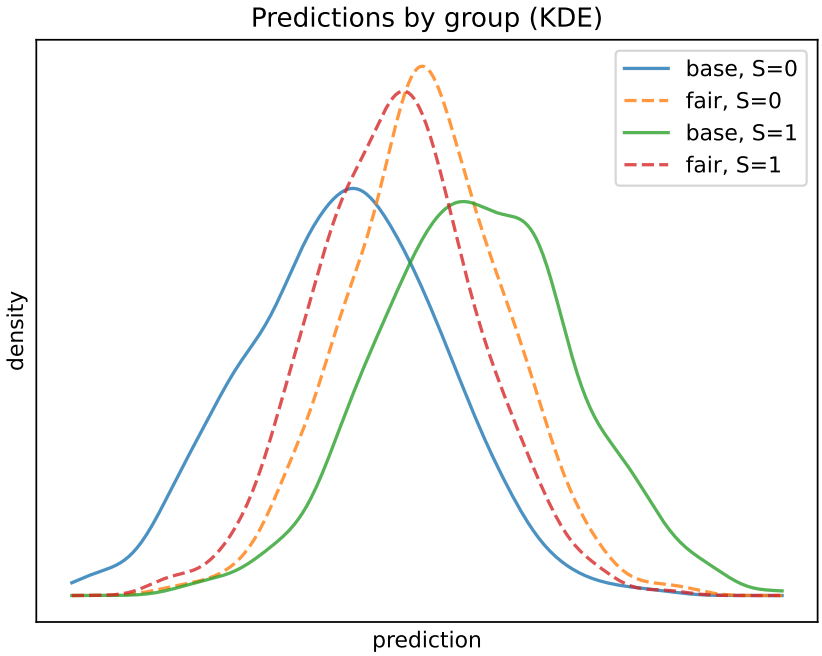
Let φ be a C-convex Kantorovich potential for $\mathrm{OT}_C(\mu_{+}^{\uparrow}, \mu_{-}^{\uparrow})$. Where $\partial_h \varphi(h, d)$ exists,

$$\boxed{f_+(h, d) = h - \frac{|d|}{2}\, \partial_h \varphi(h, d)} \qquad \text{(and symmetrically } f_-\text{)}.$$



7. Final form of the optimal fair regressor

$$f^\star(x) = \begin{cases} f_+\big(\eta(x), \Delta(x)\big), & x \in \mathcal{X}_+, \\ f_-\big(\eta(x), \Delta(x)\big), & x \in \mathcal{X}_-, \\ \eta(x), & x \in \mathcal{X}_=. \end{cases} \qquad \text{and} \qquad (f^\star)_{\#}\mu_+ = (f^\star)_{\#}\mu_- = \nu^{\text{bar}}.$$



6 Nestedness (multi to one OT)

Setup: source $(X, \mu) \subset \mathbb{R}^m$, target $(Y, \nu) \subset \mathbb{R}$, cost $c(x, y)$ differentiable in y. Define sublevel sets using the *marginal cost*

$$X_{\leq}(y, k) := \{x \in \mathcal{X} : \partial_y c(x, y) \leq k\}.$$

For each y, choose $k(y)$ so that $\mu\big(X_{\leq}(y,k(y))\big) = F_\nu(y)$ where F_ν is the CDF of ν.

Nestedness:

$$y_1 < y_2 \implies X_{\leq}\big(y_1,k(y_1)\big) \subseteq X_{\leq}\big(y_2,k(y_2)\big).$$

Consequence: existence of a Monge map $T : \mathcal{X} \to Y$ with $\partial_y c\big(x,T(x)\big) = k\big(T(x)\big)$ a.e. (no mass splitting).

Tiny example. If $c(x,y) = \frac{1}{2}\|x\|^2 - y\,x_1$, then $\partial_y c(x,y) = -x_1$, so $X_{\leq}(y,k) = \{x_1 \geq -k\}$ (independent of y). Picking $k(y)$ to match $F_\nu(y)$ makes these sets nested $\Rightarrow$ a map depending on x_1 only.

6.1 Ordering preservation between $\hat{\eta}$ (before correction) and $\tilde{y}$ (after correction).

Let n be the number of individuals, and let $\pi_{\hat{\eta}}$ (resp. $\pi_{\tilde{y}}$) denote the permutation that sorts $\hat{\eta}$ (resp. $\tilde{y}$) in increasing order. We report the following rank-based metrics.

1. **Spearman rank correlation** ρ_S.

$$\rho_S(\hat{\eta},\tilde{y}) \;=\; \mathrm{corr}\big(\mathrm{rank}(\hat{\eta}),\,\mathrm{rank}(\tilde{y})\big).$$

   Interpretation. Measures global monotonic agreement between the two rankings. $\rho_S = 1$ means perfect rank preservation; $\rho_S \approx 0$ indicates weak association; $\rho_S < 0$ indicates an inverted ranking.

2. **Kendall's** τ_b.

$$\tau_b(\hat{\eta},\tilde{y}) \;=\; \frac{C-D}{\sqrt{(C+D+T_{\hat{\eta}})(C+D+T_{\tilde{y}})}},$$

   where C (resp. D) is the number of *concordant* (resp. *discordant*) pairs (i,j), and $T_{\hat{\eta}}$ (resp. $T_{\tilde{y}}$) counts ties in $\hat{\eta}$ (resp. $\tilde{y}$).
   Interpretation. Pairwise notion of rank consistency: it estimates the fraction of pairs that keep the same order (accounting for ties). $\tau_b = 1$ means all pairs preserve their order; smaller values indicate more inversions.

3. **Mean absolute rank displacement** $\mathrm{MAD}_{\mathrm{rank}}$.

$$\mathrm{MAD}_{\mathrm{rank}}(\hat{\eta},\tilde{y}) \;=\; \frac{1}{n(n-1)}\sum_{i=1}^{n}\big|\mathrm{rank}(\hat{\eta}_i) - \mathrm{rank}(\tilde{y}_i)\big|.$$

   Interpretation. Average amount (in relative rank units) by which individuals move in the ordering after correction. Values near 0 = no reordering; values closer to 1 = large mélange.

4. **Top-k Jaccard overlap** J_k. Let $\mathrm{Top}_k(z) = \{i : \; z_i \text{ is among the } k \text{ largest entries of } z\}$.

$$J_k(\hat{\eta},\tilde{y}) \;=\; \frac{\big|\mathrm{Top}_k(\hat{\eta}) \cap \mathrm{Top}_k(\tilde{y})\big|}{\big|\mathrm{Top}_k(\hat{\eta}) \cup \mathrm{Top}_k(\tilde{y})\big|}.$$

   Interpretation. Stability of the *selected* set (e.g., admissions/shortlist): $J_k = 1$ means the same top-k individuals are selected before/after correction; $J_k = 0$ means the top-k sets are disjoint.

   $\implies$ mélange entre les groupes mais dans chaque groupe l'ordre est assez preservé

7 Classification link (cost-sensitive, $Y \in \{0,1\}$)

For weight $y \in [0,1]$,

$$R_y(g) = y\,\mathbb{P}(Y=0,\,g(X)=1) + (1-y)\,\mathbb{P}(Y=1,\,g(X)=0), \quad g : \mathcal{X} \to \{0,1\}.$$

Subset	Spearman ρ	Kendall τ_b	Mean $\lvert\Delta\text{rank}\rvert$	Top-5% Jaccard
overall	0.662	0.509	0.216	0.250
S=0	0.980	0.883	0.044	0.682
S=1	0.979	0.881	0.045	0.805

Table 1: Ordering stability between $\hat{\eta}$ and corrected scores.

Quantity	AUC
AUC$(S \mid Y)$	0.858
AUC$(S \mid \hat{\eta})$	0.828
AUC$(S \mid \tilde{y})$	0.396

Table 2: How easy it is to infer S from te score

Under DP and a *nestedness* condition, the optimal fair classifiers are *thresholds* of $f^\star$:

$$g_y^\star(x) = \mathbf{1}\{\, f^\star(x) \geq t(y)\,\},$$

and the family $\{g_y^\star\}_y$ recovers $f^\star$ (up to null sets). Without nestedness this equivalence can fail.

8 Awareness VS unawareness

- **Awareness** (can use S): DP-optimal regressor is a *groupwise* OT/quantile match to a common barycenter; preserves within-group order; often closed-form via CDF$\leftrightarrow$quantile.
- **Unawareness**: DP forces a *2D$\to$1D* OT barycenter depending on (η, Δ); no general closed form; order preservation can *fail*.

9 In practice

1. Estimate $\widehat{\eta}(x)$ and $\widehat{\Delta}(x)$ (need $\widehat{\mathbb{P}}(S = s \mid X = x)$ or density ratios).
2. Build empirical $\widehat{\mu}_\pm^{\uparrow}$ on samples from $\mathcal{X}_\pm$.
3. Solve OT_C with $C((h_1, d_1), (h_2, d_2)) = (h_2 - h_1)^2/(|d_1| + |d_2|)$ to get maps $\hat{f}_\pm$.
4. Output $\hat{f}^\star$ piecewise on $(\mathcal{X}_+, \mathcal{X}_-, \mathcal{X}_=)$ as above.

10 defs

- **DP:** $f(X) \perp S \iff f_{\#}\mu_s$ independent of s.
- **Pushforward:** $f_{\#}\mu(A) = \mu(f^{-1}(A))$.
- **Jordan decomposition:** $\alpha = \alpha^- - \alpha^+$ with disjoint supports, $|\alpha| = \alpha^+ + \alpha^-$.
- **Barycenter (here):** ν minimizing OT-distance sum from sources.
- **Nestedness:** monotone growth in y of $\partial_y c$-sublevel sets with prescribed μ-mass.
- **Kantorovich potential:** dual function whose subgradient characterizes optimal plans/maps.

11 formuless

$$\boxed{c((h, d), y) = \frac{(h - y)^2}{|d|}}, \qquad \boxed{C((h_1, d_1), (h_2, d_2)) = \frac{(h_2 - h_1)^2}{|d_1| + |d_2|}},$$

$$\boxed{f^\star(x) = \begin{cases} \eta(x) - \dfrac{|\Delta(x)|}{2}\,\partial_h\varphi(\eta(x),\Delta(x)), & x \in \mathcal{X}_+ \cup \mathcal{X}_-, \\ \eta(x), & x \in \mathcal{X}_=. \end{cases}}.$$

12 what porblem do they solve?

We want regression under **demographic parity (DP)** in the **unawareness** setting: the predictor cannot use S at inference time, but we still want $\widehat{Y} \perp S$.

Key design choice: build a *smooth convex dual objective* whose stochastic gradients can be computed from *unlabeled* X only, enabling **online post-processing** on a stream of i.i.d. X.

13 Setting and notation

Data $(X, S, Y) \in \mathbb{R}^d \times [K] \times \mathbb{R}$ with unknown law P.

- Regression function: $\eta(x) := \mathbb{E}[Y \mid X = x]$.
- Sensitive marginals: $p_s := \mathbb{P}(S = s)$, and vector $p = (p_s)_{s\in[K]}$.
- Sensitive predictor / propensity: $\tau(x) := (\tau_s(x))_{s\in[K]}$ with $\tau_s(x) := \mathbb{P}(S = s \mid X = x)$.

Randomized predictors (unawareness). A randomized prediction function is a Markov kernel

$$\pi : \begin{cases} \mathcal{B}(\mathbb{R}) \times \mathbb{R}^d & \to [0,1], \\ B & \mapsto \pi(B \mid x). \end{cases}$$

It induces a prediction random variable $\widehat{Y}_\pi$ by

$$Law(\widehat{Y}_\pi \mid X = x, S = s) = \pi(\cdot \mid x).$$

Remark 2.1: deterministic predictors correspond to Dirac kernels $\pi(\cdot \mid x) = \delta_{g(x)}$, giving $\widehat{Y}_\pi = g(X)$. Also, π specifies the joint law of $(X, S, \widehat{Y}_\pi)$, and to relate it to (X, S, Y) one may assume conditional independence $\widehat{Y}_\pi \perp Y \mid (X, S)$ so that $(X, S, Y, \widehat{Y}_\pi)$ is well-defined.

14 Demographic parity and risk

DP. A predictor π satisfies demographic parity if

$$\widehat{Y}_\pi \perp S.$$

This is *distributional*, not pointwise in x.

Risk. They measure performance against η (excess risk):

$$R(\pi) := \mathbb{E}\big[(\widehat{Y}_\pi - \eta(X))^2\big] = \mathbb{E}\Big[\int_{\mathbb{R}} (\widehat{y} - \eta(X))^2 \, \pi(d\widehat{y} \mid X)\Big].$$

15 discretization of the prediction range

They force the predictor outputs to lie on a **fixed finite grid**.

Bounded signal assumption. Assume there exists $B > 0$ such that $|\eta(X)| \le B$ a.s.

Grid. For an integer $L \ge 1$, define the uniform grid on $[-B, B]$

$$\widehat{\mathcal{Y}}_L := \left\{ \widehat{y}_\ell := \frac{\ell B}{L} : \ell \in [[L]] := \{-L, \dots, L\} \right\}, \qquad |\widehat{\mathcal{Y}}_L| = 2L + 1.$$

A feasible predictor must satisfy $Supp(\pi(\cdot \mid x)) = \widehat{\mathcal{Y}}_L$ for all x.

Why discretize? Because with a fixed finite support, DP reduces to finitely many *linear* constraints on the probabilities $\pi(\ell \mid x)$:

$$\mathbb{P}(\widehat{Y}_\pi = \widehat{y}_\ell \mid S = s) = \mathbb{P}(\widehat{Y}_\pi = \widehat{y}_\ell) \quad \Longleftrightarrow \quad \mathbb{E}[\pi(\ell \mid X) \mid S = s] = \mathbb{E}[\pi(\ell \mid X)].$$

They allow slack via thresholds $\varepsilon = (\varepsilon_s)_{s\in[K]}$ and define (per group, per bin)

$$U_s(\pi, \ell) := \Big|\mathbb{E}[\pi(\ell \mid X) \mid S = s] - \mathbb{E}[\pi(\ell \mid X)]\Big| \le \varepsilon_s.$$

Important sanity point (parce que j'ai eu un doute): this is *not* turning regression into classification. The algorithm learns a *distribution* over grid values; you can output either a sample $\widehat{Y}_\pi$ or the mean $\mathbb{E}[\widehat{Y}_\pi \mid X = x] = \sum_\ell \widehat{y}_\ell\, \pi(\ell \mid x)$, which is still a real-valued regressor. The discretization is an algorithmic device to make DP constraints finite.

16 Entropic regularization ⇒ smooth dual convex program

Solving the discretized constrained problem directly is a large LP in dimension $\Omega(LK)$, so they switch to an entropic-regularized variant (comme c'est leur gros argument je pense que dans la comparaison il faudra voir la complexité temporelle *a minima*).

Entropy-regularized risk. Let $\Psi(\mu) = \sum_{y\in Supp(\mu)} \mu(y) \log \mu(y)$ denote negative entropy. For $\beta > 0$:

$$R_\beta(\pi) := R(\pi) + \frac{1}{\beta}\mathbb{E}\big[\Psi(\pi(\cdot \mid X))\big].$$

Dual variables and key reparameterization. Define

$$t(x) := \mathbf{1} - \frac{\tau(x)}{p} \in \mathbb{R}^K \qquad \text{(componentwise division)}, \qquad r_\ell(x) := \big(\eta(x) - \widehat{y}_\ell\big)^2.$$

16.1 Lemma 3.1 (the main structural lemma)

The entropic-regularized problem admits a solution that can be written explicitly in terms of nonnegative dual matrices $\Lambda = (\lambda_{\ell s})$ and $V = (\nu_{\ell s})$, as follows. 2 matrices parce que contrainte en valeur absolue

Dual objective (smooth convex). Let $\lambda_\ell = (\lambda_{\ell s})_{s\in[K]}$ and $\nu_\ell = (\nu_{\ell s})_{s\in[K]}$. Define the log-sum-exp at temperature β by

$$\mathrm{LSE}_\beta(w) = \beta^{-1} \log\Big(\sum_j e^{\beta w_j}\Big),$$

and the soft-argmax / softmax by $\sigma_j(w) = \frac{e^{w_j}}{\sum_i e^{w_i}}$. Then define

$$F(\Lambda, V) := \mathbb{E}\Big[\mathrm{LSE}_\beta\Big(\big(\langle \lambda_\ell - \nu_\ell,\, t(X)\rangle - r_\ell(X)\big)_{\ell\in[[L]]}\Big)\Big] \;+\; \sum_{\ell\in[[L]]} \langle \lambda_\ell + \nu_\ell, \varepsilon\rangle, \quad \Lambda, V \ge 0.$$

Primal solution (softmax form). If $(\Lambda^\star, V^\star)$ minimizes F, then an optimal predictor is

$$\boxed{\pi_{\Lambda^\star, V^\star}(\ell \mid x) = \sigma_\ell\Big(\beta\big(\langle \lambda^\star_{\ell'} - \nu^\star_{\ell'},\, t(x)\rangle - r_{\ell'}(x)\big)_{\ell'\in[[L]]}\Big)\,.}$$

Interpretation. When $\Lambda = V = 0$, (no fairness constraints), we have

$$\pi_{0,0}(\ell \mid x) \propto \exp\big(-\beta(\eta(x) - \widehat{y}_\ell)^2\big),$$

so the distribution concentrates around the grid point closest to $\eta(x)$; the dual terms then reweight bins to enforce DP across groups.

16.2 Regularity + why gradient norm matters (Lemma 3.4 + 3.5)

Let

$$\sigma^2 := 2 \sum_{s \in [K]} \frac{1 - p_s}{p_s}.$$

Then ∇F is $(\beta\sigma^2)$-Lipschitz, so larger β makes F less smooth.

Since the dual domain is constrained ($\Lambda, V \geq 0$), they use the **gradient mapping** (for stepsize $\alpha > 0$)

$$G_\alpha(\Lambda, V) := \frac{(\Lambda, V) - \big((\Lambda, V) - \alpha \nabla F(\Lambda, V)\big)_+}{\alpha},$$

where $(\cdot)_+$ is coordinatewise projection onto $\mathbb{R}_+^{2(2L+1)K}$.

Lemma 3.5 (key implication). For any $(\Lambda, V) \geq 0$, the *fairness violation* of $\pi_{\Lambda,V}$ is controlled by $\|G_\alpha(\Lambda, V)\|$, and the *risk suboptimality* is bounded by a term linear in $\|G_\alpha(\Lambda, V)\|$ plus the entropic bias $\log|\widehat{\mathcal{Y}}_L|/\beta$.

Moral: they need algorithms that directly control $\mathbb{E}\|G_\alpha\|^2$ with clean statistical rates.

17 Algorithm: online unlabeled post-processing (Algorithm 1)

Inputs: L, β, number of stochastic gradient calls T, p, B, and functions η, τ.

1. Build grid $\widehat{\mathcal{Y}}_L$.
2. Define $F(\Lambda, V)$ as above; set smoothness proxy $M = \beta\sigma^2$.
3. Run a black-box stochastic optimizer $\mathcal{A}$ on F for T stochastic gradients to get $(\widehat{\Lambda}, \widehat{V})$.
4. Output $\pi_{\widehat{\Lambda},\widehat{V}}$ via the softmax formula.

Stochastic gradient oracle uses only X. They derive ∇F explicitly and estimate it by sampling a single $X \sim P_X$ and removing the expectation. So the post-processing step can be done on a **stream** of unlabeled X without storing data.

Unknown η and τ. Train $\widehat{\eta}$ and $\widehat{\tau}$ on a labeled dataset, then run the same post-processing on an independent unlabeled stream; additional error terms appear that depend on the quality of $\widehat{\eta}, \widehat{\tau}$.

18 Main guarantees (Theorem 5.1 + plugin extension)

They choose parameters as functions of the number of unlabeled samples/iterations T (e.g. $L \asymp \sqrt{T}$ and a matching β schedule) and prove that:

- fairness violation decreases at a rate essentially $1/\sqrt{T}$ (up to logs and constants),
- excess risk (vs. a natural deterministic DP benchmark based on Kolmogorov–Smirnov-type constraints) also decreases like $1/\sqrt{T}$,
- dependence on K appears through σ and on constraint tightness through $\|(\Lambda^\star, V^\star)\|$.

They also provide post-processing-style bounds for the plug-in version with $(\widehat{\eta}, \widehat{\tau})$, separating estimation error from optimization/post-processing error.

19 what is actually run

Datasets:

- Law School: predict GPA (scaled to $[0, 1]$),
- Communities & Crime: predict violent crime rate (scaled to $[0, 1]$),

with ethnicity as sensitive attribute (binary: white vs non-white).

Pipeline: split into train / unlabeled / test (0.4 / 0.4 / 0.2); train $\widehat{\eta}$ via LinearRegression and $\widehat{\tau}$ via LogisticRegression; run Algorithm 1 using the unlabeled split; evaluate risk and unfairness vs iterations.

They compare against Agarwal–Dudik–Wu (2019)in two training regimes (with and without access to unlabeled data) $\implies$ competitive or better tradeoffs.

20 Formules

- Grid indices: $[[L]] = \{-L, \dots, L\}$, grid values $\widehat{y}_\ell = \ell B/L$.
- Slack DP constraints: $U_s(\pi, \ell) \le \varepsilon_s$.
- Dual objective: $F(\Lambda, V) = \mathbb{E}[\mathrm{LSE}_\beta(\cdots)] + \sum_\ell \langle \lambda_\ell + \nu_\ell, \varepsilon \rangle$.
- Output predictor: $\pi_{\Lambda,V}(\ell \mid x) = \mathrm{softmax}_\ell\big(\beta(\langle \lambda_{\ell'} - \nu_{\ell'}, t(x) \rangle - r_{\ell'}(x))\big)$.
- Gradient mapping: $G_\alpha(\Lambda, V) = \frac{(\Lambda,V) - ((\Lambda,V) - \alpha \nabla F)_+}{\alpha}$.

21 Comparison: OT characterization vs online convex post-processing

One-line takeaway

- **Divol–Gaucher:** population-level characterization of the optimal DP regressor via an OT barycenter on a lifted space (η, Δ) (mainly two groups).
- **Taturyan–Chzhen–Hebiri:** practical post-processing algorithm with finite-sample bounds, based on discretization + entropic smoothing + stochastic convex optimization, usable with an unlabeled stream of X.

Differences

	OT	**Unlabeled post-processing**
Primary goal	Characterize $f^\star$ solving $\min_f \mathbb{E}[(Y - f(X))^2]$ s.t. $f(X) \perp S$ (population optimum).	IMPLEMENTABLE estimator achieving (approx.) DP with rates, using post-processing.
Groups	Focuses on $S \in \{1, 2\}$ (extension to $K > 2$ left for future work).	Handles $S \in [K]$ and discusses multiple protected attributes.
Form of predictor	Deterministic regressor $f^\star : \mathcal{X} \to \mathbb{R}$, described via OT on a lift $(\eta(x), \Delta(x))$.	Randomized predictor $\pi(\cdot \mid x)$ supported on a **grid** $\widehat{\mathcal{Y}}_L$ (can output a sample or the mean).
How DP is enforced	**Exact** DP constraint at the population level ($f(X) \perp S$).	**Relaxed / discretized** DP constraints on grid atoms (with slack ε); plus entropic regularization.
Key mathematical object	OT barycenter with a *multi-to-one* cost after lifting; DP governed by "disagreement" regions and a signed density gap Δ.	Smooth convex dual objective $F(\Lambda, V)$; optimal π is an explicit **softmax** (Lemma 3.1).
What information is implicitly needed	Needs $\eta(x)$ and $\Delta(x)$, where Δ depends on group-membership probabilities $\mathbb{P}(S = s \mid X = x)$ (implicit use of S).	Needs $\eta(x)$ and $\tau(x) = (\mathbb{P}(S = s \mid X = x))_s$ to form $t(x) = \mathbf{1} - \tau(x)/p$.
Computation / "online"	typically offline	Stochastic convex optimization with an oracle computable from *unlabeled* i.i.d. samples of X; suited for streaming.

Fainress through $\Delta(x)$ or $\tau(x)$.

- **OT characterization** more interpretable: what the optimal $f^\star$ looks like, why order-preservation can fail, and how DP reshapes predictions geometrically.
- **unlabeled post-processing** more applicable *pipeline*: a concrete algorithm with controllable approximation (L, β, ε) and finite-sample/optimization guarantees.

$\implies$ interesting to see the statiqsticak guarantees we can have with the OT post-processing + see what happens with $K > 2$.

Main difference in implementation:

- OT: we learn the transport on the same data we trained on they use the X_unlab to